

Water: bent molecules and C_{2v}

Water Point group C_{2v}

Z rotation: off Reset camera Reset operation

Symmetry operations Settings

Symmetry operations

Choose an operation · click again to replay

4 operations · select one to play

identity · 1

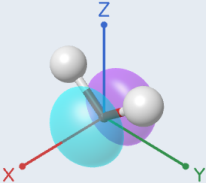
E

proper rotation · 1

$C_2(z)$

reflection · 2

$\sigma_v(xz)$ $\sigma_v(yz)$



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Ammonia: a threefold principal axis

Ammonia Point group C_{3v}

Z rotation: off

Reset camera

Reset operation

Symmetry operations

Settings

Symmetry operations

Choose an operation · click again to replay

6 operations · select one to play

identity · 1

E

proper rotation · 2

$C_3(+120^\circ)$

$C_3(-120^\circ)$

reflection · 3

$\sigma_v(1)$

$\sigma_v(2)$

$\sigma_v(3)$

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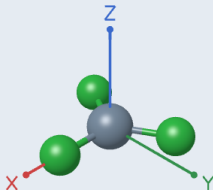
Boron trifluoride: planar D_{3h} symmetry

Boron trifluoride Point group D_{3h}

Z rotation: off

Reset camera

Reset operation



Symmetry operations

Settings

Symmetry operations

Choose an operation · click again to replay

12 operations · select one to play

identity · 1

E

proper rotation · 5

$C_3(+120^\circ)$

$C_3(-120^\circ)$

$C_2'(1)$

$C_2'(2)$

$C_2'(3)$

reflection · 4

$\sigma_h(xy)$

$\sigma_v(1)$

$\sigma_v(2)$

$\sigma_v(3)$

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Ethylene: inversion and three mirror planes

Ethylene Point group **D_{2h}**

Z rotation: off Reset camera Reset operation

Symmetry operations Settings

Symmetry operations
Choose an operation · click again to replay
8 operations · select one to play

identity · 1
E

proper rotation · 3
 $C_2(x)$ $C_2(y)$ $C_2(z)$

reflection · 3
 $\sigma(xy)$ $\sigma(xz)$ $\sigma(yz)$

inversion · 1

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Methane: tetrahedral symmetry

Methane Point group T_d

Z rotation: off

Reset camera

Reset operation

Symmetry operations

Settings

Symmetry operations

Choose an operation · click again to replay

24 operations · select one to play

identity · 1

E

proper rotation · 11

$C_3(1, +120^\circ)$

$C_3(1, -120^\circ)$

$C_3(2, +120^\circ)$

$C_3(2, -120^\circ)$

$C_3(3, +120^\circ)$

$C_3(3, -120^\circ)$

$C_2(4, +120^\circ)$

$C_2(4, -120^\circ)$

$C_2(x)$

$C_2(y)$

$C_2(z)$

reflection · 6

$\sigma^d(x-y)$

$\sigma^d(x+y)$

$\sigma^d(x-z)$

$\sigma^d(x+z)$

$\sigma^d(y-z)$

$\sigma^d(y+z)$

improper rotation · 6

$S_6(x, +90^\circ)$

$S_6(x, -90^\circ)$

$S_6(y, +90^\circ)$

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